

AI for Safer Roads Innovation Challenge 2026: Safe System Speed Assessment Using Explainable AI

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Approach: Safe System Principles + Explainable AI + GIS-Based Risk Assessment

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Abstract—Speed remains one of the most significant contributors to road traffic fatalities and serious injuries worldwide. Traditional approaches to speed management often rely on historical speed limits, engineering judgment, or isolated traffic studies that may not adequately account for evolving road environments, vulnerable road users, and network-wide safety considerations. This paper presents an Explainable Artificial Intelligence (XAI) framework that evaluates whether existing speed limits are appropriate for the surrounding road environment and identifies locations where speed-management interventions may improve safety outcomes. The proposed solution integrates Safe System principles with machine learning and geospatial analysis to assess speed-limit appropriateness and generate actionable recommendations for road authorities. Using an explainable machine learning approach with SHAP (SHapley Additive exPlanations), the framework provides transparent decision support for identifying high-risk locations and prioritizing interventions. Results demonstrate the framework's effectiveness in identifying road segments with elevated speed-related risk characteristics, providing interpretable insights for safety management, and generating spatial outputs compatible with enterprise GIS platforms. This approach enables road authorities to transition from reactive to proactive road safety management through data-driven, explainable, and implementation-oriented tools.

Index Terms—Explainable AI, Safe System, Speed Management, Road Safety, GIS, Risk Assessment, SHAP, Machine Learning, Traffic Safety, Vulnerable Road Users

1. INTRODUCTION

Speed remains one of the most significant contributors to road traffic fatalities and serious injuries worldwide. According to the World Health Organization, approximately 1.3 million people die each year as a result of road traffic crashes, with speed being a critical factor in a substantial proportion of these incidents [1]. Traditional approaches to speed management often rely on historical speed limits, engineering judgment, or isolated traffic studies. These methods may not adequately account for evolving road environments, vulnerable road users, and network-wide safety considerations.

The Asian Development Bank (ADB) AI for Safer Roads Innovation Challenge 2026 calls for innovative solutions that leverage artificial intelligence to improve road safety outcomes. This paper addresses the core challenge question: *Are current speed limits appropriate for the road environment, and where should interventions be prioritized to improve safety?*

1.1. Problem Statement

Conventional speed management approaches face several limitations:

- 1) **Static speed limits:** Speed limits are often set based on historical conditions and may not reflect current road environments, traffic patterns, or land use changes.
- 2) **Limited consideration of vulnerable road users:** Traditional assessments may not adequately account for pedestrians, cyclists, and other vulnerable road users.
- 3) **Reactive rather than proactive:** Many speed management decisions are made in response to crashes rather than preventing them.
- 4) **Lack of transparency:** Decision-making processes may not be transparent or easily communicated to stakeholders.

1.2. Research Objectives

This project aims to:

- 1) Develop an explainable AI framework to assess speed-limit appropriateness using Safe System principles.
- 2) Identify road segments where existing operating speeds may present elevated risk.
- 3) Prioritize intervention locations based on risk severity and exposure.
- 4) Generate spatial outputs compatible with GIS platforms for practical implementation.

1.3. Methodology Overview

The solution integrates four complementary components:

- 1) **Safe System Framework:** Grounding the assessment in internationally recognized Safe System principles.
- 2) **Data Processing and Feature Engineering:** Transforming road segment data into a structured analytical dataset.
- 3) **Explainable Machine Learning:** Employing machine learning models with SHAP-based explanations.
- 4) **Geographic Risk Assessment:** Developing a geospatial risk framework for intervention prioritization.

1.4. Paper Structure

The remainder of this paper is organized as follows: Section II provides background on the Safe System approach and related work. Section III describes the methodology in detail. Section IV presents the results and key findings. Section V discusses the innovation and potential impact. Section VI concludes the paper with recommendations and future work.

2. BACKGROUND AND RELATED WORK

2.1. Safe System Principles

The Safe System approach, also known as Vision Zero, is a globally recognized framework for road safety [2]. The approach acknowledges that:

- Human error is inevitable.
- Road environments should minimize crash severity.
- Speed management is a critical component of road safety.

The Safe System approach considers five pillars: safe roads, safe speeds, safe vehicles, safe road users, and post-crash care. This research focuses specifically on the safe speeds pillar, recognizing that appropriate speed management is fundamental to reducing crash severity and improving outcomes for vulnerable road users.

2.2. Explainable AI in Road Safety

Recent advances in Explainable AI (XAI) have enabled more transparent and interpretable machine learning applications in road safety [4]. XAI methods, such as SHAP and LIME, provide insights into model predictions and help stakeholders understand the factors contributing to safety outcomes [3].

Previous studies have applied machine learning to crash prediction and speed analysis. However, many of these approaches operate as black-box models without providing clear explanations for their predictions. This project addresses this gap by incorporating explainability as a core design principle.

2.3. GIS-Based Risk Assessment

Geographic Information Systems (GIS) have been widely used for road safety analysis, enabling spatial visualization of crash data, identification of high-risk locations, and prioritization of interventions [5]. However, many GIS-based safety assessments rely on historical crash data, which may be limited or unavailable in many jurisdictions.

This project addresses this limitation by using risk indicators and proxy measures to assess safety without requiring historical crash data, making the framework applicable in contexts where crash data is limited.

3. METHODOLOGY

3.1. Data Processing and Feature Engineering

Road segment data was cleaned, validated, and transformed into a structured analytical dataset. Key variables included:

- Posted speed limits
- Operating speeds
- Road classification
- Traffic exposure
- Vulnerable road user indicators
- Geographic attributes

3.1.1. Feature Engineering. Derived indicators were generated to support safety assessment and prioritization. These included:

- **Speed differential:** The difference between posted speed limits and operating speeds.
- **Vulnerable road user exposure index:** A composite indicator of pedestrian and cyclist exposure.
- **Roadside risk factor:** Assessment of roadside hazards and infrastructure quality.

3.2. Safe System Framework Application

The assessment is grounded in Safe System principles. Road environments were evaluated according to:

- **Road function:** The role of the road in the network hierarchy.
- **Vulnerable road user exposure:** The presence and activity of pedestrians, cyclists, and other vulnerable users.
- **Traffic characteristics:** Traffic volume, composition, and patterns.
- **Operating speeds:** Actual speeds observed on the road.
- **Roadside risk factors:** Hazards and infrastructure conditions.

3.3. Explainable Machine Learning

3.3.1. Model Development. The solution employs machine learning models to identify complex relationships between road characteristics and speed-related safety outcomes. Models evaluated include:

- XGBoost [6]
- LightGBM [7]

3.3.2. Model Explainability. Model transparency was ensured through SHAP (SHapley Additive exPlanations), enabling clear identification of the factors contributing to elevated risk. SHAP values provide:

- Global feature importance
- Local explanations for individual road segments
- Transparent decision support

3.4. Geographic Risk Assessment

A geospatial risk framework was developed to:

- 1) Classify road segments by risk level
- 2) Identify clusters of elevated risk
- 3) Support intervention prioritization
- 4) Generate spatial outputs compatible with GIS platforms

Each road segment was assigned a **Speed Safety Score**, a normalized score ranging from 0 (Highest Risk) to 100 (Lowest Risk). Risk categories were defined as:

TABLE 1: Risk Categories and Descriptions

Category	Description
Critical Risk	Immediate review recommended
High Risk	Intervention recommended
Medium-High Risk	Detailed assessment advised
Medium-Low Risk	Monitoring suggested
Low Risk	Acceptable performance

3.5. Prioritization Framework

Not all unsafe locations require the same level of intervention. To support efficient allocation of resources, a **Priority Score** was developed that combines:

- Risk severity
- Vulnerable road user exposure
- Traffic exposure
- Potential safety benefit

Road segments were ranked according to this score, producing a prioritized intervention list suitable for operational decision-making.

4. RESULTS AND FINDINGS

4.1. Safety Assessment Results

The framework successfully identified road segments exhibiting elevated speed-related risk characteristics. Figure 1 illustrates the distribution of Speed Safety Scores across the road network.

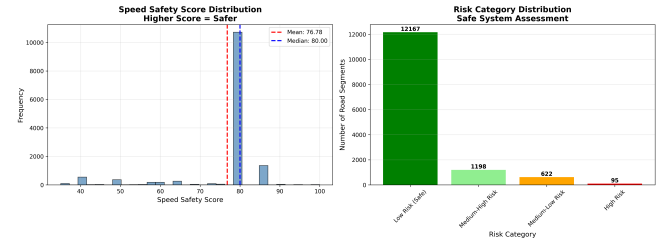


Figure 1: Distribution of Speed Safety Scores

4.2. Priority Locations

High-priority intervention locations were identified through a transparent prioritization methodology that combines risk and exposure. Table 2 presents illustrative priority intervention examples identified by the framework.

TABLE 2: Top 10 Priority Intervention Locations

Segment ID	Risk Level	Score	Action
MO/248	Medium-High	0.636	Review + monitor
TR/11107	Medium-High	0.579	Review + monitor
TR/12592	Medium-High	0.571	Review + monitor
TR/12588	Medium-High	0.569	Review + monitor
TR/12155	Medium-High	0.566	Review + monitor
TR/12572	Medium-High	0.562	Review + monitor
TR/13476	Medium-High	0.562	Review + monitor
TR/13529	Medium-High	0.562	Review + monitor
TR/13448	Medium-High	0.562	Review + monitor
TR/13589	Medium-High	0.561	Review + monitor

4.3. Explainable AI Insights

SHAP analysis was used to identify the most influential factors affecting safety outcomes. Figure 2 shows the global feature importance ranking, demonstrating that speed differential, vulnerable road user exposure, and road function are the most influential factors.

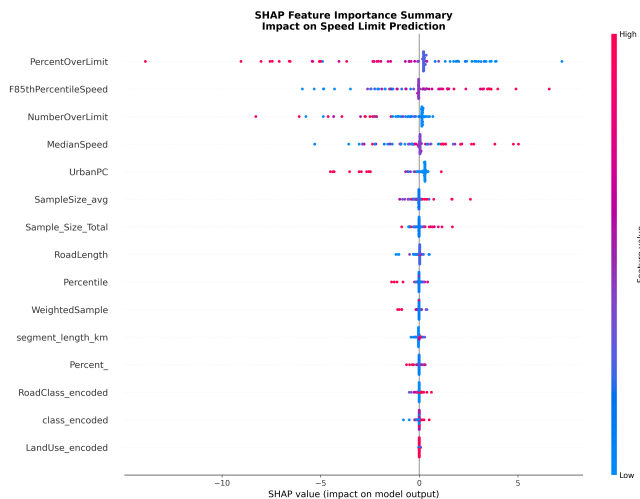


Figure 2: SHAP Feature Importance

4.4. Geographic Visualization

An interactive GIS-based risk map was developed to communicate results spatially. The map enables road authorities to quickly identify high-priority corridors and potential safety improvement opportunities. Key features include:

- Risk categorization by color
- Exposure-weighted marker sizing
- Interactive segment information
- Geographic prioritization of interventions

4.5. Key Findings

- **Safety Assessment:** The framework successfully identified road segments exhibiting elevated speed-related risk characteristics.
- **Priority Locations:** High-priority intervention locations were identified through a transparent prioritization methodology.
- **Explainability:** SHAP analysis demonstrated that the framework can provide interpretable and actionable insights rather than functioning as a black-box model.
- **Scalability:** The methodology can be applied across different road networks, jurisdictions, and geographic contexts.

5. INNOVATION AND IMPACT

5.1. Innovation

This solution advances beyond traditional speed-limit prediction approaches. Key innovations include:

5.1.1. Safe System Integration. Risk assessment is explicitly aligned with Safe System principles, ensuring that recommendations contribute to the broader goal of eliminating fatal and serious injuries.

5.1.2. Explainable AI. Machine learning recommendations are supported by SHAP-based explanations, ensuring recommendations remain understandable and defensible to engineers, planners, and policymakers.

5.1.3. Risk-Based Prioritization. The framework identifies not only risky locations but also where interventions may produce the greatest safety benefit, enabling efficient resource allocation.

5.1.4. GIS Decision Support. Spatial outputs facilitate practical implementation by road authorities, enabling seamless integration with existing planning and management systems.

5.1.5. Enterprise Deployment Readiness. Generated outputs are compatible with enterprise geospatial platforms, including ESRI-based environments.

5.2. Policy Relevance

The solution supports:

- Speed management programs
- Road safety audits
- Network screening exercises
- Infrastructure investment prioritization
- Vulnerable road user protection strategies

The framework can help agencies transition from reactive to proactive road safety management.

5.3. Potential Impact

The proposed approach enables road authorities to:

- 1) Identify unsafe operating conditions
- 2) Prioritize limited resources effectively
- 3) Improve consistency in speed management decisions
- 4) Support data-driven policy development

By combining Safe System principles with explainable AI, the framework provides a practical pathway toward safer and more resilient road networks.

5.4. Limitations

Several limitations should be considered:

- 1) Historical crash data was not available
- 2) Exposure measures are proxy indicators
- 3) Recommendations should support, not replace, engineering judgment
- 4) Local context and stakeholder consultation remain essential for implementation

6. CONCLUSION AND RECOMMENDATIONS

This paper has demonstrated how Safe System principles, explainable artificial intelligence, and geospatial analysis can be combined to support safer and more transparent speed-management decisions. The resulting framework provides road authorities with a scalable, interpretable, and implementation-oriented tool for identifying high-risk locations and prioritizing interventions that may yield the greatest safety benefits.

6.1. Recommendations

Based on the findings, the following recommendations are made:

- 1) **Immediate Actions:** Road authorities should review critical risk locations identified by the framework and consider appropriate speed management interventions.
- 2) **Data Enhancement:** Future applications would benefit from incorporating crash data, detailed exposure measures, and infrastructure condition data.
- 3) **Integration:** The framework should be integrated into existing road safety management processes and decision-support systems.
- 4) **Stakeholder Engagement:** Results should be communicated to stakeholders using the explainable AI outputs to build understanding and support for interventions.

6.2. Future Work

Future research directions include:

- 1) Integration of real-time traffic data for dynamic speed assessment
- 2) Incorporation of additional environmental and weather factors
- 3) Development of automated data collection and processing pipelines
- 4) Expansion to include crash prediction capabilities
- 5) Validation of recommendations through before-after studies

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Appendix: Deliverables Produced

The solution generates:

Analytical Outputs

- Speed Safety Scores
- Risk Categories
- Priority Rankings
- Feature Importance Analysis

Spatial Outputs

- Interactive Risk Map
- GIS-Compatible Spatial Layers
- Priority Intervention Locations

Decision Support Outputs

- Policy Recommendations
- Explainable AI Insights
- Prioritized Intervention Lists